

SIMATS ENGINEERING ASSESSMENT TOOL

COURSE NAME: Theory of Computation

COURSE CODE: CSA13

COURSE OUTCOME COVERED

CO2: Design Finite Automata DFA, NFA, ϵ -NFA and demonstrate their equivalence for recognizing regular languages. (BL:3)

Assessment Tool Weightage: 40%

QUESTIONS: 5
Duration : 1 Hour

Total Marks:25

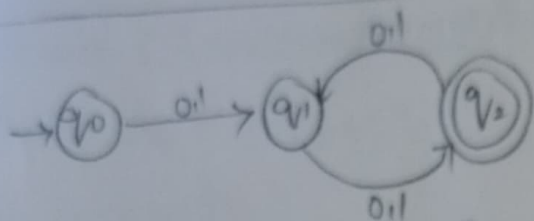
Design Task

Code of Conduct:

I Shaik Hafeeza Tarannum
192372014 (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.

I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student:



Regular Expression for the Given Automata:

From the diagram.

$\Rightarrow q_0$ is initial state

$\star q_2$ is Final state

$\cdot q_0 \xrightarrow{0} q_1$

$\cdot q_1 \xrightarrow{1} q_2$

$\cdot q_2 \xrightarrow{0} q_1$

To reach the final state, the input must contain an Even number of symbols, with at least 2 symbols.

Regular Expression:

$(0+1)(0+1)(0+1)(0+1)^*$

Equivalent form:

$((0+1)(0+1))^+$

Language:

$L = \{ w \in \{0,1\}^* \mid |w| \text{ is even and } |w| \geq 2 \}$

Ex:

00, 01, 10, 11, 0000, 0101, 1010, ...

(2A)

Given Grammar

The grammar is

$S \rightarrow aSc \mid T U \mid \epsilon$

$T \rightarrow bTc \mid \epsilon$

$U \rightarrow aUb \mid \epsilon$

A Language defined by the Grammar

For T

$$T \Rightarrow b^n c^n$$

Using $S \rightarrow aSc$ we get

$$a^m b^n c^{n+m}$$

For U:

$$U \Rightarrow a^n b^m$$

Using $S \rightarrow aSc$, we get

$$a^{m+n} b^m c^m$$

$$\therefore L = \{a^m b^n c^{m+n} \mid m, n \geq 0\} \cup \{a^{m+n} b^m c^m \mid m, n \geq 0\}$$

(B) Derivation Tree for $aaabbc$.

$$S \Rightarrow aSc, S \Rightarrow U$$

$$S \Rightarrow aUc, U \Rightarrow aUb$$

$$S \Rightarrow aaUbc, U \Rightarrow aUb$$

$$S \Rightarrow aaaUbc, U \Rightarrow \epsilon$$

$$S \Rightarrow aaabbc.$$

Derivation Tree



$\therefore aaabbc$ is left-most Tree

(C) Is the Grammar ambiguous?

Yes, the Grammar is ambiguous.

Ex, consider It can be generated both T and U.

Through T.

$$S \Rightarrow aSc, S \Rightarrow T$$

$$S \Rightarrow aTc,$$

$$S \Rightarrow ac,$$

Using U

$$S \Rightarrow aSc \Rightarrow aUc \Rightarrow ac.$$

$\therefore$ Thus the same string ac has two different derivations.

$\therefore$ The grammar is ambiguous.

NFA with ϵ Transitions for $00^*1 + 11^*0$.

Given. $R \in = 00^*1 + 11^*0$.

This represents:

$\Rightarrow 00^*1$: one or more 0's followed by 1

$\Rightarrow 11^*0$: one or more 1's followed by 0.

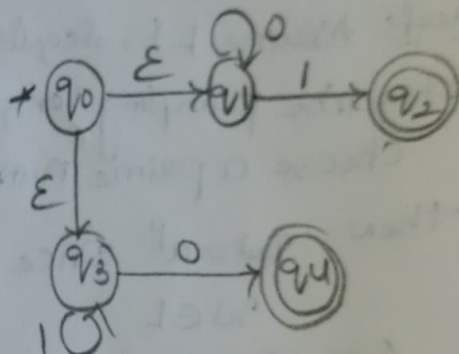
NFA

Use q_0 as the initial state

Transition Table

State	ϵ	Next state
q_0	ϵ	q_1
q_0	ϵ	q_3
q_1	0	q_1
q_1	1	q_2 *
q_3	1	q_3
q_3	0	q_4 *

State diagram



Initial state = q_0

Final state = q_2, q_4

So the NFA accepts

$00^*1 + 11^*0$

Example accepted: 001, 01, 0001, 10, 110, 1110.

(4) (i) If L_1 and L_2 are regular, then $L_1 + L_2$ is also regular.

Given L_1, L_2 are regular.

Since regular languages are closed under union.

$L_1 \cup L_2$ is regular

Explanation: If M_1 recognizes L_1 and M_2 recognizes L_2 , we can construct an NFA with a new ^{initial} state and ϵ -transitions to the initial states of M_1 and M_2 .

$\therefore L_1 \cup L_2$

Hence, $L_1 \cup L_2$ is regular.

(i) $L = \{a^p \mid p \text{ is prime}\}$ regular?

No, it is not regular.

Proof: Assume L is regular.

then n be pumping length.

Choose a prime number, $p > n$.

Then $w = a^p$ since p is prime:

$w \in L$

According to pumping Lemma:

$w = xyz$

where: $|xy| \leq n, |y| > 0$

Let: $|y| = k, k \geq 1$.

pump y ($p+1$) times: $xy^{p+1}z$

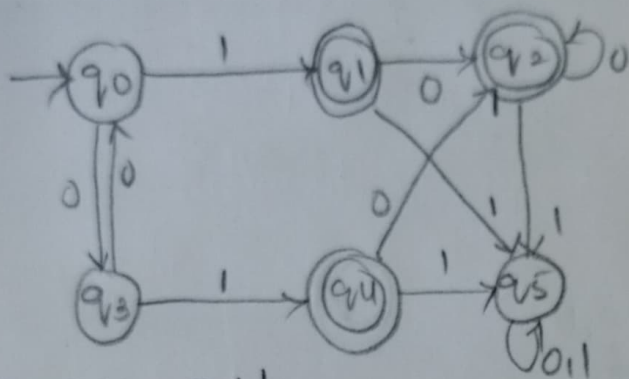
Its length becomes: $p + pk \Rightarrow p(1+k)$

Since $p > 1$, and $1+k > 1$, the number $p(1+k)$ is composite
 not prime

$\therefore xy^{p+1}z \notin L$

Hence, $\{a^p \mid p \text{ is prime}\}$ is not regular.

Minimization of DFA



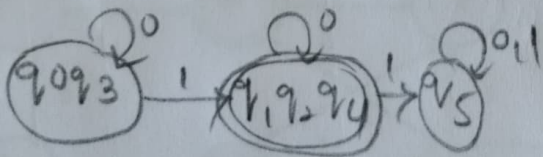
Transition Table:

State	0	1
→ q ₀	q ₃	q ₁
* q ₁	q ₂	q ₅
* q ₂	q ₂	q ₅
q ₃	q ₀	q ₄
* q ₄	q ₂	q ₅
q ₅	q ₅	q ₅

Step 1: Cared states into Final and Non-Final.

$\{q_0, q_3, q_5\}$ (Non-F) $\{q_1, q_2, q_4\}$ (F)

$\{q_0, q_3\}$ & $\{q_5\}$ & $\{q_1, q_2, q_4\}$



Here $A = \{q_0, q_3\}$

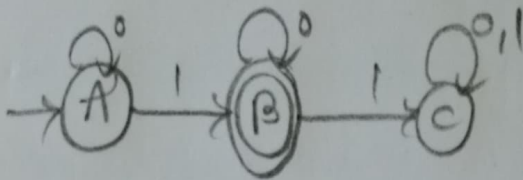
$B = \{q_1, q_2, q_4\}$

$C = \{q_5\}$

Transition Minimized

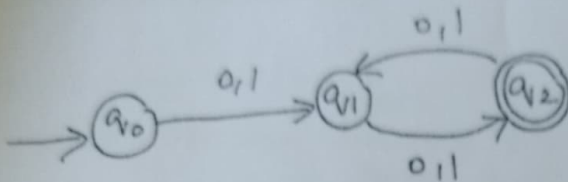
<u>S</u>	0	1
→ A	A	B
* B	B	C
C	C	C

State diagram Minimized



SIMATS ENGINEERING ASSESSMENT TOOL

1. Design the regular expression for the language represented by the given automata



2. Consider the following grammar

$S \rightarrow aSc|T|U|e$

$T \rightarrow bTc|e$

$U \rightarrow aUb|e$

A) What is the language defined by the grammar?

B) Give the derivation tree for the word aaabbc

C) A Grammar is Ambiguous if there exists two distinct derivations for a word. Is the above grammar ambiguous? Justify.

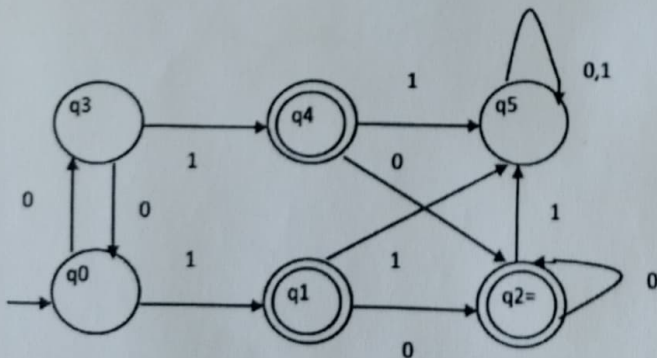
3. Design a non deterministic finite automata with ϵ -transitions to represent the language denoted by the regular expression

00^*1+11^*0

4. i) Show that if L_1 and L_2 are regular then (L_1+L_2) also regular

ii) Is the language $\{a^p \mid p \text{ is prime}\}$ regular? Justify

5. Design minimized automata, Consider the DFA given below. Find the states that are equivalent and construct a minimum state automata equivalent to the given automata. Draw the minimized automata, write its transition table and mark initial and final states



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RUBRICS FOR CALCULATION

Criteria	Weightage	Level 4 – Exemplary (5 Marks)	Level 3 – Proficient (4 Marks)	Level 2 – Developing (2–3 Marks)	Level 1 – Beginning (0–1 Mark)
Conceptual Understanding	30	Demonstrates clear and complete understanding of the concept	Good understanding with minor gaps	Basic understanding with some misconceptions	Poor or incorrect understanding
Accuracy of Answer	25	Completely correct answer with no errors	Mostly correct with minor errors	Partially correct with noticeable errors	Incorrect or irrelevant answer
Application of Knowledge	20	Correctly applies concepts to solve the question/problem	Applies concepts with minor mistakes	Limited or partially correct application	No or incorrect application
Completeness of Response	15	Covers all required parts of the question	Covers most parts with minor omissions	Covers only some parts	Incomplete or missing response
Clarity & Presentation	10	Clear, concise, and well-organized answer	Understandable with minor issues	Somewhat unclear or poorly structured	Difficult to understand

96/100